

Chapter 2. Randomized Experiments and Stats Review

Now that you know the basics about causality, its time to talk about the inference part in causal inference. This chapter will first recap some concepts from the previous chapter in the context of randomized experiments. Randomized experiments are the gold standard for causal inference, so it is really important that you understand what makes them special. Even when randomization is not an option, having it as an ideal to strive for will be immensely helpful when thinking about causality.

Next, I'll use randomized experiments to review some important statistical concepts and tools, such as error, confidence interval, hypothesis tests, power, and sample size calculations. If you know about all of this, I'll make it clear when the review will start so you can skip it.

Brute-Force Independence with Randomization

In the previous chapter, you saw why and how association is different from causation. You also saw what is required to make association equal to causation:

$$E[Y|T = 1] - E[Y|T = 0] = \underbrace{E[Y_1 - Y_0|T = 1]}_{ATT} + \underbrace{\{E[Y_0|T = 1] - E[Y_0|T = 0]\}}_{BIAS}$$

To recap, association can be described as the sum of two components: the average treatment effect on the treated and the bias. The

measured association is only fully attributed to causation if the bias component is zero. There will be no bias if $E[Y_t|T = 0] = E[Y_t|T = 1]$. In other words, association will be causation if the treated and control are equal or comparable, except for their treatment. Or, in slightly more technical terms, when the potential outcomes of the treated are equal to the potential outcomes of the untreated, at least in expectations. Remember that potential outcome Y_{ti} is the outcome you would see had unit i received treatment t .

In [Chapter 1](#), I also briefly touched on how to equate association and causation in the case that the potential outcomes are independent from the treatment:

$$(Y_0, Y_1) \perp T.$$

Importantly, I'm *not* talking about the independence between the treatment and the outcome. If that were the case, the treatment would have no impact on the outcome for you to measure. For example, let's say that the treatment is a new feature in your company's app and the outcome is time spent in that app. Saying that $Feature \perp TimeSpent$ means that time spent in the app is the same in both treated and untreated groups. In other words, the new feature has simply no effect.

Instead, what you want is for the *potential outcomes* to be independent of the treatment. There is an important distinction here. Saying that $Y_1 \perp T$ means that the outcome that would have been observed had the subjects received the treatment is independent of whether they actually received it or not. Analogously, $Y_0 \perp T$ indicates that the outcome that would have been observed if the subjects were untreated does not depend on the actual treatment

assignment. In summary, the outcome Y that was actually observed still depends on the treatments actually assigned.

Another simpler way of putting this is that the independence assumption implies that treatment and control groups are comparable. Or that knowing the treatment assignment doesn't give me any information about the baseline potential outcome, Y_0 . Consequently, $(Y_0, Y_1) \perp T$ means that the treatment is the only thing that causes difference in outcome between treatment and control:

$$E[Y_0|T = 0] = E[Y_0|T = 1] = E[Y_0]$$

and

$$E[Y_1|T = 0] = E[Y_1|T = 1] = E[Y_1]$$

which, as you've seen, allows a simple comparison between treated and control averages to identify the ATE:

$$E[Y|T = 1] - E[Y|T = 0] = E[Y_1 - Y_0] = ATE$$

Although independence is nothing more than an assumption, you can make it a lot more plausible if you randomize the treatment T . By doing so, you are tying the treatment assignment to the flip of a coin; that is, a random mechanism completely known to us. This coin doesn't have to be fair. You could assign the treatment to just 10% of the subjects, 1%, or even less. As long as the assignment mechanism is random, you can get the right conditions to identify the treatment effect.

By randomizing the treatment, you ensure that the treated and control group are roughly (in expectation) comparable. The only systematic difference between them is the treatment itself, which allows you to attribute any difference in the outcome to that treatment. Essentially,

randomization brute-forces your way toward independence between the treatment and the *potential* outcomes.

Let's now take all this math and go over an example so you can see that it is actually quite simple. In the following section, I'll use randomized control trials (RCT) in order to understand the impact of cross-sell emails.

An A/B Testing Example

A common strategy among companies is to have a cheap or even free product that isn't profitable, but serves as the doorway to attracting new customers. Once the company has those customers, it can then cross-sell other products that are more profitable. Let's suppose you work for a coffee delivery company. Your main product is a low-cost monthly subscription that allows the customers to have high-quality and curated coffee delivered to them weekly. Beyond this basic and low-cost subscription, your company provides a more premium one, with brewing perks and the world's finest coffee, like that from local producers in the small town of Divinolandia, Brazil. This is by far your most profitable service, and therefore your goal is to increase its sales to the users who have already subscribed for your low-cost, entry product. For that, your company has a marketing team that tries to sell the premium coffee delivery subscription to its customers. They do this mainly through cross-sell emails. As the causal inference expert, your goal is to understand how effective those emails are.

When you look into the existing data (nonrandomized) to answer this question, you can clearly see that customers who received an email were more likely to buy the premium subscription. In technical terms, when a customer buys the product you are trying to sell, you can say

they converted. So, you can say that the customers who received an email converted more:

$$E[\text{Conversion} | \text{Email} = 1] > E[\text{Conversion} | \text{Email} = 0]$$

Unfortunately, you also discover that the marketing team tends to send emails for the customers who they thought were more likely to convert in the first place. It is not entirely clear how they did this. Maybe they looked for customers who interacted the most with the company, or those who answered positively in a satisfaction survey. Regardless, this is very strong evidence that

$$E[\text{Conversion}_0 | \text{Email} = 1] > E[\text{Conversion}_0 | \text{Email} = 0].$$

In other words, customers who were actually sent the email would convert in greater numbers than other customers, even if no email was sent at all. As a result, a simple comparison in means is a biased estimator of the true causal effect of the cross-sell email. To solve that, you need to make the treated and untreated comparable:

$E[Y_0 | T = 1] = E[Y_0 | T = 0]$, which can be done by randomly assigning the emails. If you manage to do that, the treated and untreated will have, on average, the same conversion, except for the treatment they receive. So, suppose you did just that. You selected three random samples from your customer base. To one of them, you didn't send any emails. To the other, you sent a large and beautifully written email about the premium subscription. To the last sample, you sent a short and to-the-point email about the premium subscription. After some time collecting data, you have something that looks like this:

```
In [1]: import pandas as pd # for data manipulation
        import numpy as np # for numerical computation

        data = pd.read_csv("../data/cross_sell_email.csv")
```

data

	gender	cross_sell_email	age
0	0	short	15
1	1	short	27
...	...	...	...
321	1	no_email	16
322	1	long	24

323 rows × 4 columns

You can see that you have 323 observations. It's not exactly big data, but something you can work with.

SIMULATED VERSUS REAL-WORLD DATA

When teaching about causal inference, it is very helpful to use simulated data. First, because causal inference is always accompanied by a statement about how the data was generated. Simulation allows me to talk about this assignment mechanism without any uncertainty. Second, causal inference involves counterfactual quantities that I can choose to show in order to give a better explanation of what is going on. However, so that the data doesn't look too artificial, I often take real-world data and transform it to fit the example I'm trying to give. For instance, this example takes data from the paper "A Randomized Assessment of Online Learning" (2016), by William T. Alpert et al., and transforms it to look like cross-sell email data.

To estimate the causal effect, you can simply compute the average conversion for each of the treatment groups:

```
In [2]: (data
         .groupby(["cross_sell_email"])
         .mean())
```

	gender	age	conversion
cross_sell_email			
long	0.550459	21.752294	0.055046
no_email	0.542553	20.489362	0.042553
short	0.633333	20.991667	0.125000

Yup. It's really that simple. You can see that the group assigned to no email had a conversion rate of 4.2%, while the groups assigned to the long and short email had a conversion rate of 5.5% and a whopping 12.5%, respectively. Therefore, the ATEs, measured as the difference between each treated group and the control group,

$ATE = E[Y|T = t] - E[Y|T = 0]$, where 1.3 and 8.3 percentage points increase for the long and short email, respectively.

Interestingly, sending an email that is short and to the point seems better than an elaborated one.

The beauty of RCTs is that you no longer have to worry if the marketing team somehow targeted customers who were likely to convert or, for that matter, you don't have to worry that the customers from the distinct treatment groups are different in any systematic way, other than the treatment they received. By design, the random experiment is made to wipe out those differences, making $(Y_0, Y_1) \perp T$, at least in theory.

In practice, a good sanity check to see if the randomization was done right (or if you are looking at the correct data) is to check if the treated are equal to the untreated in pretreatment variables. For example, you have data on gender and age and you can see whether these two characteristics are balanced across treatment groups.

When you look at age, treatment groups seem very much alike, but there seems to be a difference in gender (*woman* = 1, *man* = 0). It seems that the group that received the short email had 63% men, compared to 54% in the control group and 55% in the group that got the long email. This is somewhat unsettling, as the treatment group in which you found the highest impact also appears to be different from the other groups. So, even if independence should hold in theory in RCTs, it does not necessarily hold in practice. It could be that the large effect you saw for the short email was due to the fact that, for whatever reason, $E[Y_0|man] > E[Y_0|woman]$.

There isn't a clear consensus on how to evaluate balance, but one very simple suggestion is to check the normalized differences between the treatment groups:

$$\frac{\hat{\mu}_{tr} - \hat{\mu}_{co}}{\sqrt{(\hat{\sigma}_{tr}^2 + \hat{\sigma}_{co}^2)/2}},$$

where $\hat{\mu}$, $\hat{\sigma}^2$ are the sample mean and variance, respectively. Since there are three treatment groups in your example, you can just compute this difference with respect to the control group:

```
In [3]: X = ["gender", "age"]

mu = data.groupby("cross_sell_email")[X].mean()
var = data.groupby("cross_sell_email")[X].var()

norm_diff = ((mu - mu.loc["no_email"]) /
              np.sqrt((var + var.loc["no_email"])))

norm_diff
```

	gender	age
cross_sell_email		
long	0.015802	0.221423
no_email	0.000000	0.000000
short	0.184341	0.087370

If this difference is too small or too large, you should be worried. Unfortunately, there isn't a clear threshold for how much difference is too much, but 0.5 seems to be a good rule of thumb. In this example, you don't have any difference that is that high, but it does seem that

the group that got the short email has a large difference in gender, while the group that got the long email has a large difference in age.

SEE ALSO

For a more in-depth discussion of this topic, check out section 14.2 of the book *Causal Inference for Statistics, Social, and Biomedical Sciences: An Introduction*, by Guido W. Imbens and Donald B. Rubin (Cambridge University Press).

If the preceding formula seems a bit magic right now, don't worry. It will become clearer once you go over the statistical review part of this chapter. For now, I just want to draw your attention to what happens with a small dataset. Even under randomization, it could be that, by chance, one group is different from another. In large samples, this difference tends to disappear. It also brings forth the issue of how much difference is enough for you to conclude that the treatments are indeed effective and not just due to chance, which is something I'll address shortly.

The Ideal Experiment

Randomized experiments or randomized controlled trials are the most reliable way to get causal effects. It's a straightforward technique and absurdly convincing. It is so powerful that most countries have it as a requirement for showing the effectiveness of new drugs. Think of it this way: if you could, RCT would be all you would ever do to uncover causality. A well-designed RCT is the dream of any scientist and decision maker.

Unfortunately, they tend to be either very expensive—both in money, but more importantly, in time—or just plain unethical. Sometimes, you simply can't control the assignment mechanism. Imagine yourself as a physician trying to estimate the effect of smoking during pregnancy

on baby weight at birth. You can't simply force a random portion of moms to smoke during pregnancy. Or say you work for a big bank, and you need to estimate the impact of the credit line on customer churn. It would be too expensive to give random credit lines to your customers. Or say you want to understand the impact of increasing the minimum wage on unemployment. You can't simply assign countries to have one or another minimum wage. Moreover, as you will see in [Chapter 3](#), there are some situations (selection biased ones) where not even RCTs can save you.

Still, I would like you to think about random experiments beyond a tool for uncovering causal effects. Rather, the goal here is to use it as a benchmark. Whenever you do causal inference without RCTs, you should always ask yourself what would be the perfect experiment to answer your question. Even if that ideal experiment is not feasible, it serves as a valuable benchmark. It often sheds some light on how you can discover the causal effect even without such an experiment.

The Most Dangerous Equation

Now that you understand the value of an experiment, it's time to review what it means to not have infinite data. Causal inference is a two-step process. RCTs are invaluable in helping with identification, but if the sample size of an experiment is small, you'll struggle with the second step: inference. To understand this, it's worth reviewing some statistical concepts and tools. If you are already familiar with them, feel free to skip to the next chapter.

In his famous article of 2007, Howard Wainer writes about very dangerous equations:

“Some equations are dangerous if you know them, and others are dangerous if you do not. The first category may pose danger because the secrets within its bounds open doors behind which lies terrible peril. The obvious winner in this is Einstein’s iconic equation $E = MC^2$, for it provides a measure of the enormous energy hidden within ordinary matter. [...] Instead I am interested in equations that unleash their danger not when we know about them, but rather when we do not. Kept close at hand, these equations allow us to understand things clearly, but their absence leaves us dangerously ignorant.”

The equation he talks about is Moivre’s equation:

$$SE = \frac{\sigma}{\sqrt{n}}$$

where SE is the standard error of the mean, σ is the standard deviation, and n is the sample size. This math is definitely something you should master, so let’s get to it.

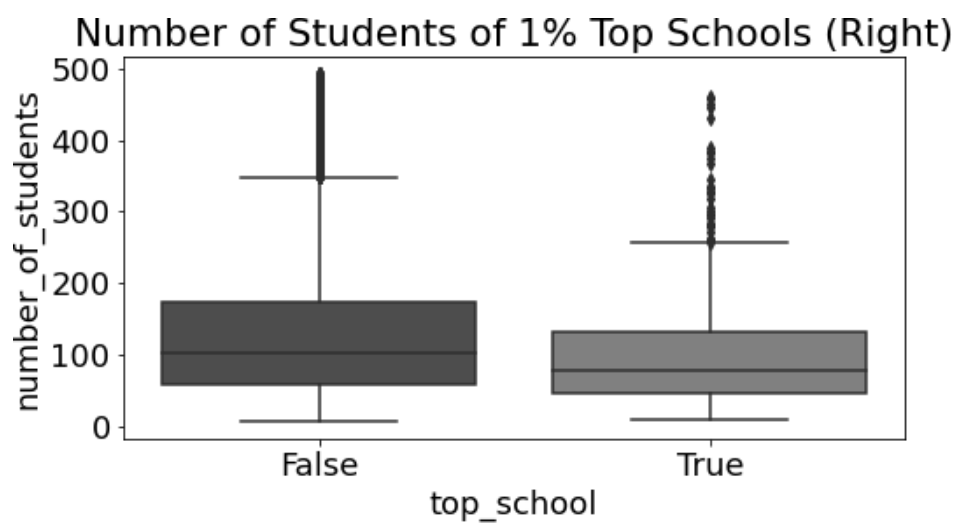
To see why not knowing this equation is very dangerous, let’s look at some education data. I’ve compiled data on ENEM scores (Brazilian standardized high school scores, similar to SATs) from different schools over a three-year period. I’ve also cleaned the data to keep only the information relevant to you in this section.

If you look at the top-performing school, something catches the eye—those schools have a reasonably small number of students:

```
In [4]: df = pd.read_csv("data/enem_scores.csv")
        df.sort_values(by="avg_score", ascending=False)
```

	year	school_id	number_of_stude
16670	2007	33062633	68
16796	2007	33065403	172
...	...	...	...
14636	2007	31311723	222
17318	2007	33087679	210

Looking at it from another angle, you can separate only the 1% of top schools and study them. What are they like? Perhaps you can learn something from the best and replicate it elsewhere. And sure enough, if you look at the top 1% of schools, you'll figure out they have, on average, fewer students:



One natural conclusion is that small schools lead to higher academic performance. This makes intuitive sense, since we believe that fewer students per teacher allows the teacher to give focused attention to each student. But what does this have to do with Moivre's equation? And why is it dangerous?

Well, it becomes dangerous once people start to make important and expensive decisions based on this information. In his article, Howard continues:

In the 1990s, it became popular to champion reductions in the size of schools. Numerous philanthropic organizations and government agencies funded the division of larger schools because students at small schools are overrepresented in groups with high test scores.

What people forgot to do was to also look at the bottom 1% of schools: they also have very few students!

What you see in [Figure 2-1](#) is precisely what's expected according to Moivre's equation. As the number of students grows, the average score becomes more and more precise. Schools with very few students (low sample size) can have very high and low scores simply due to chance. This is less likely to occur in large schools. Moivre's equation talks about a fundamental fact regarding the reality of information and records in the form of data: it is always imprecise. The question then becomes: how imprecise? And what can you do to take those inaccuracies into account?

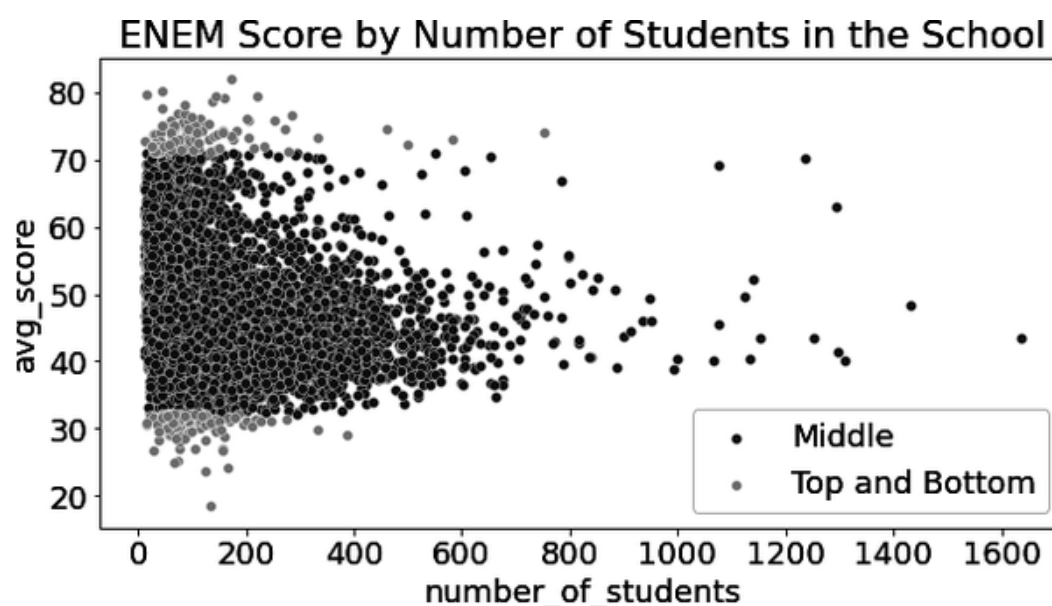


Figure 2-1. A typical triangular plot showing how variance decreases with sample size

One way to quantify our uncertainty is the *variance of our estimates*. Variance tells you how much observation deviates from its central (expected) value. As Moivre's equation indicates, this uncertainty shrinks as the amount of data you observe increases. This makes sense, right? If you see many students performing excellently at a school, you can be more confident that this is indeed a good school. However, if you see a school with only 10 students and 8 of them perform well, you need to be more suspicious. By chance, it could be that the school got some above-average students.

The beautiful triangular plot you see in [Figure 2-1](#) tells precisely this story. It shows you how your estimate of the school performance has a huge variance when the sample size is small. It also indicates that variance shrinks as the sample size increases. This is true for the average score in a school, but it is also true about any summary statistics you might have, including the ATE you often want to estimate. Back to our cross-sell email application, if you had thousands of customers in each treatment group, instead of hundreds, you would

be much more confident that the difference in conversion you saw between treated and control groups are not simply due to chance.

RANDOM AND SYSTEMATIC ERROR

Another way to think about this uncertainty in the data is to *contrast systematic error with random error*. Systematic errors are consistent biases that affect all measurements in the same way, while random errors are unpredictable fluctuations in data due to chance. Systematic error, or bias, doesn't diminish as you gather more data, as it pushes all measurements to the same direction, away from the quantity you want to estimate. In contrast, random error decreases as the sample size increases, as seen in Moivre's equation. Statistics is the science that deals with these imprecisions due to random error, so they don't catch you off-guard. It's a way to take uncertainty into account.

The Standard Error of Our Estimates

Since this is just a review of statistics, I'll take the liberty to go a bit faster. If you are not familiar with distributions, variance, and standard errors, please read on, but keep in mind that you might need some additional resources. I suggest you google any MIT course on introduction to statistics. They are usually quite good and you can watch them for free on YouTube.

In ["The Most Dangerous Equation"](#), you estimated the average treatment effect $E[Y_1 - Y_0]$ as the difference in the means between the treated and the untreated $E[Y|T = 1] - E[Y|T = 0]$. Specifically, you figured out the ATE for two types of cross-sell emails on conversion. You then saw that the short email had a very impressive lift, of more than 8 percentage points, while the long email had a smaller impact, of just 1.3 percentage points increase. But there is still a lingering question: are those effects large enough so you can be confident they are not due to chance? In technical terms, do you know if they are statistically significant?

To do so, you first need to estimate the SE , according to the equation I've shown earlier. n is pretty easy to get. You just need the `len` of each treatment. Or, you can use pandas `groupby` followed by a `size` aggregation:

```
In [5]: data = pd.read_csv("../data/cross_sell_email.csv")

short_email = data.query("cross_sell_email=='short_email'")
long_email = data.query("cross_sell_email=='long_email'")
email = data.query("cross_sell_email!='no_email'")
no_email = data.query("cross_sell_email=='no_email'")

data.groupby("cross_sell_email").size()
```

```
Out[5]: cross_sell_email
      long          109
      no_email       94
      short         120
      dtype: int64
```

To get the estimate for the standard deviation, you can apply the following equation:

$$\hat{\sigma} = \sqrt{\frac{1}{N-1} \sum_{i=0}^N (x - \bar{x})^2}$$

where $\bar{x}$ is the mean of x .

HATS

In this book, I'll use hats to denote the sample estimate of parameters and predictions.

Fortunately for you, most programming software already implements this. In pandas, you can use the method `std`. Putting it all together, you

have the following function for the standard error:

```
In [6]: def se(y: pd.Series):  
        return y.std() / np.sqrt(len(y))  
  
        print("SE for Long Email:", se(long_email))  
        print("SE for Short Email:", se(short_email))
```

```
Out[6]: SE for Long Email: 0.021946024609185506  
        SE for Short Email: 0.030316953129541618
```

Knowing this formula is incredibly handy (we'll come back to it multiple times, trust me), but know that pandas also has a built-in method for calculating the standard error, `.sem()` (as in standard error of the mean):

```
In [7]: print("SE for Long Email:", long_email.sem())  
        print("SE for Short Email:", short_email.sem())
```

```
Out[7]: SE for Long Email: 0.021946024609185506  
        SE for Short Email: 0.030316953129541618
```

Confidence Intervals

The standard error of your estimate is a measure of confidence. You need to go into turbulent and polemical statistical waters to understand precisely what it means. For one view of statistics, the frequentist view, we would say that our data is nothing more than a manifestation of an underlying data-generating process. This process is abstract and ideal. It is governed by true parameters that are unchanging but also unknown to us. In the context of cross-sell email, if you could run multiple experiments and calculate the conversion rate for each of them, they would fall around the true underlying conversion rate, even though not being exactly equal to it. This is very much like Plato's writing on the Forms:

Each [of the essential forms] manifests itself in a great variety of combinations, with actions, with material things, and with one another, and each seems to be many.

To understand this, let's suppose you have the true abstract distribution of conversion for the short cross-sell email. Because conversion is either zero or one, it follows a Bernoulli distribution and let's say that the probability of success in this distribution is 0.08. That is, whenever a customer receives the short email, it has an 8% chance of converting. Next, let's pretend you can run 10,000 experiments. On each one, you collect a sample of 100 customers, send them the short email and observe the average conversion, giving you a total of 10,000 conversion rates. The 10,000 conversion rates from those experiments will be distributed around the true mean of 0.08 (see [Figure 2-2](#)). Some experiments will have a conversion rate lower than the true one, and some will be higher, but the mean of the 10,000 conversion rate will be pretty close to the true mean.

LIST COMPREHENSION

I tend to use a lot of list comprehension instead of `for` loops whenever I want to apply a function to every item in a sequence. A list comprehension is just a syntactic sugar for a mapping `for` loop:

```
table_2 = []
for n in range(11):
    table_2.append(n*2)

table_2 = [n*2 for n in range(11)]
```

```
In [8]: n = 100
        conv_rate = 0.08

        def run_experiment():
            return np.random.binomial(1, conv_rate, size=n)

        np.random.seed(42)

        experiments = [run_experiment().mean() for _ in range(1000)]
```

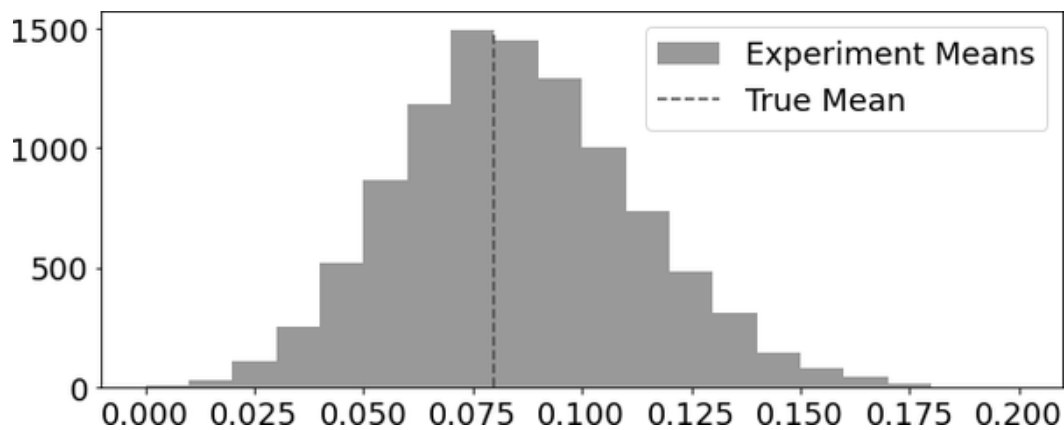
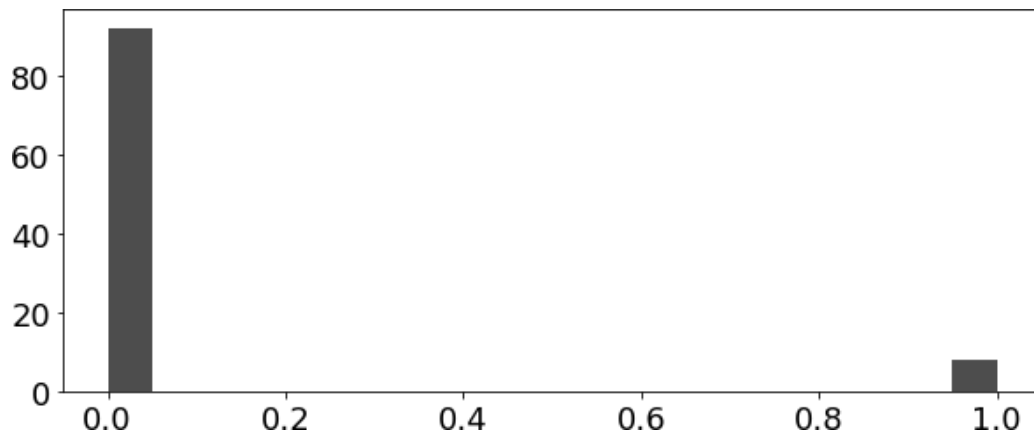


Figure 2-2. The distribution of conversion rate (average conversion) of 10,000 experiments, each with 100 units

This is to say that you can never be sure that the mean of your experiment matches the true platonic and ideal mean. However, *with the standard error, you can create an interval that will contain the true mean in 95% of the experiments you run.*

In real life, you don't have the luxury of simulating the same experiment with multiple datasets. You often only have one. But you can draw from the idea of simulating multiple experiments to construct a *confidence interval*. Confidence intervals come with a probability attached to them. The most common one is 95%. This probability tells you that if you were to run multiple experiments and construct the 95% confidence interval in each one of them, the true mean would fall inside the interval 95% of the time.

To calculate the confidence interval, you'll use what is perhaps the most mind-blowing result in statistics: the Central Limit Theorem. Take a closer look at the distribution of conversion rates you've just plotted. Now, remember that conversion is either zero or one and hence follows a Bernoulli distribution. If you plot this Bernoulli distribution in a histogram, it will have a huge bar at 0 and small bar at 1, since the success rate is only 8%. This looks nothing like a normal distribution, right?



This is where that mind-blowing result comes into play. Even though the distribution of the data is not normally distributed (like in the conversion case, which follows a Bernoulli distribution), the *average of the data is always normally distributed*. If you collect data on conversion multiple times and calculate the average conversion each time, those averages will follow a normal distribution. This is very neat, because normal distribution is well known and you can do all sorts of interesting things with it. For example, for the purpose of calculating the confidence interval, you can leverage knowledge from statistical theory that 95% of the mass of a normal distribution falls between 2 standard deviations (see [Figure 2-3](#)) above and below the mean (technically, 1.96, but 2 is a good approximation that is easier to remember).

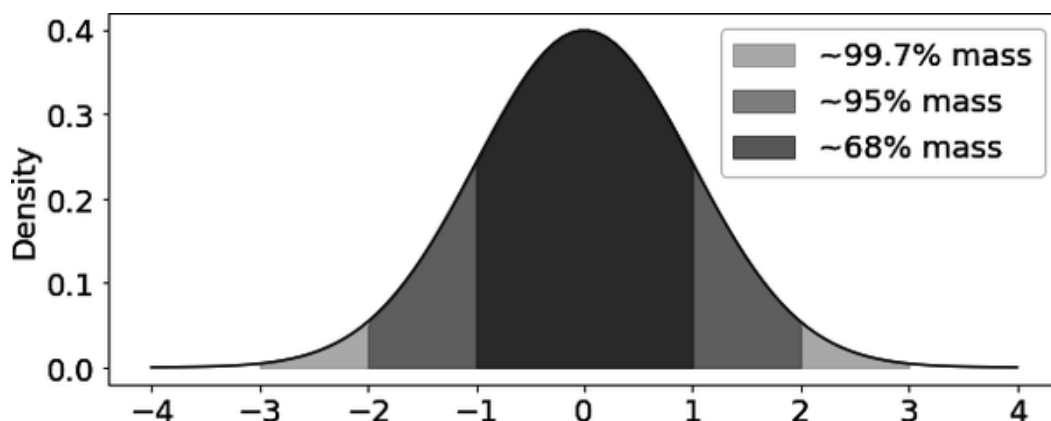
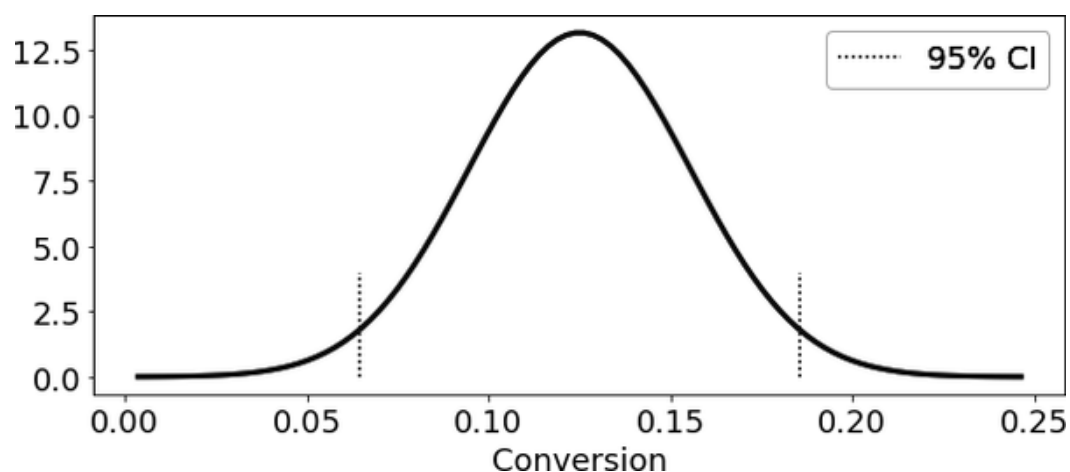


Figure 2-3. Standard normal distribution

Back to your cross-sell experiments, you now know that the conversion rate would follow a normal distribution, if you could run multiple similar experiments. The best estimate you have for the mean of that (unknown) distribution is the mean from your small experiment. Moreover, the standard error serves as your estimate of the standard deviation of that unknown distribution for the sample mean. So, if you multiply the standard error by 2 and add and subtract it from the mean of your experiments, you will construct a 95% confidence interval for the true mean:

```
In [9]: exp_se = short_email.sem()
        exp_mu = short_email.mean()
        ci = (exp_mu - 2 * exp_se, exp_mu + 2 * exp_se)
        print("95% CI for Short Email: ", ci)
```

```
Out[9]: 95% CI for Short Email: (0.06436609374091676,
```



Of course, you don't need to restrict yourself to the 95% confidence interval. If you want to be more careful you could generate the 99% interval instead. You just need to multiply the standard deviation by the factor that will contain 99% of the mass of a normal distribution.

To find that factor, you can use the `ppf` function from `scipy`. This function gives you the inverse of cumulative distribution function (CDF) of a standard normal distribution. For example, `ppf(0.5)` will return 0.0, saying that 50% of the mass of the standard normal distribution is below 0.0. So, for any significance level α , the factor you need to multiply the *SE* by in order to get a $1 - \alpha$ confidence interval is given by $|ppf((1 - \alpha)/2)|$:

```
In [10]: from scipy import stats

        z = np.abs(stats.norm.ppf((1-.99)/2))
        print(z)
        ci = (exp_mu - z * exp_se, exp_mu + z * exp_se)
        ci
```

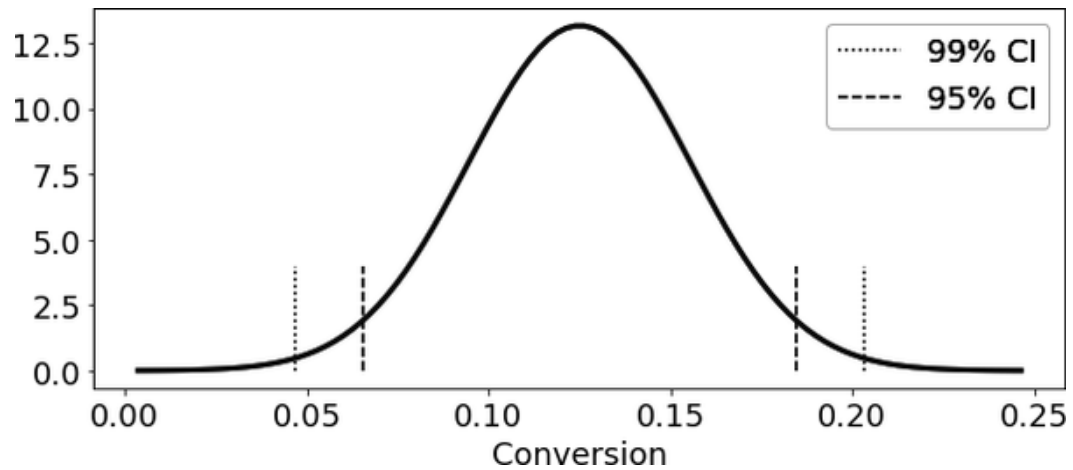
```
Out[10]: 2.5758293035489004
```

```
Out[10]: (0.04690870373460816, 0.20309129626539185)
```

```
In [11]: stats.norm.ppf((1-.99)/2)
```



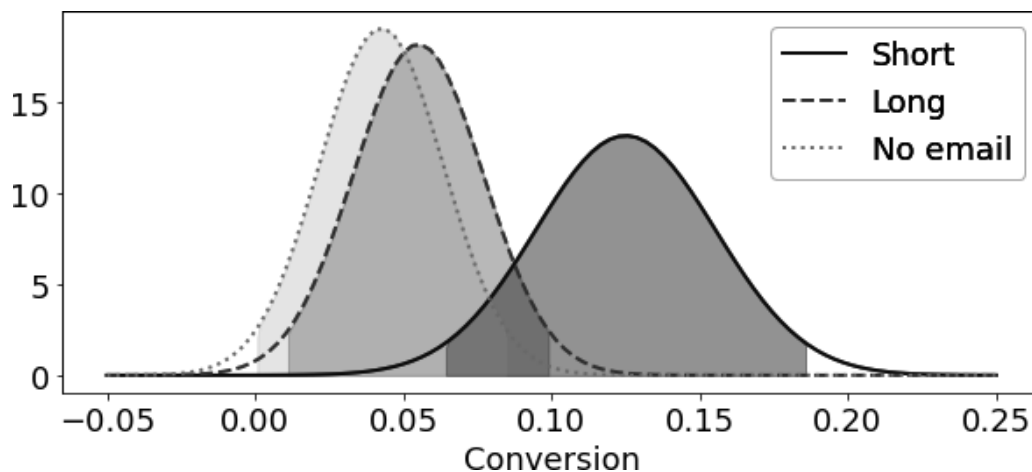
```
Out[11]: -2.5758293035489004
```



That is for the short email. You could also show the 95% CI for the conversion rate associated with the other treatment groups:

```
In [12]: def ci(y: pd.Series):  
         return (y.mean() - 2 * y.sem(), y.mean() + 2 * y.sem())  
  
print("95% CI for Short Email:", ci(short_email))  
print("95% CI for Long Email:", ci(long_email))  
print("95% CI for No Email:", ci(no_email))
```

```
Out[12]: 95% CI for Short Email: (0.06436609374091676,  
          0.20113390625908324)  
95% CI for Long Email: (0.01115382234126202,  
          0.11115382234126202)  
95% CI for No Email: (0.0006919679286838468,  
          0.0006919679286838468)
```



Here, you can see that the 95% CI of the three groups overlap with each other. If they didn't, you would be able to conclude that the difference in conversion between the groups is not simply by chance. In other words, you would be able to say that sending a cross-sell email causes a statistically significant difference in conversion rates. But since the intervals do overlap, you can't say that. At least not yet. Importantly, overlapping confidence intervals is not enough to say that the difference between the groups is not statistically significant, however, if they didn't overlap, that would mean they are statistically different. In other words, nonoverlapping confidence intervals is conservative evidence for statistical significance.

To recap, confidence intervals are a way to place uncertainty around your estimates. The smaller the sample size, the larger the standard error, and hence, the wider the confidence interval. Since they are super easy to compute, lack of confidence intervals signals either some bad intentions or simply lack of knowledge, which is equally concerning. Finally, you should always be suspicious of measurements without any uncertainty metric.

THE EFFECTIVENESS OF COVID-19 VACCINES

Randomized control trials are incredibly important for the pharmaceutical industry. Perhaps the most widely known examples are the tests conducted to determine the effectiveness of COVID-19 vaccines, given the tremendous impact those had on almost everyone on the planet. Here is the result section from the study *Efficacy and Safety of the mRNA-1273 SARS-CoV-2 Vaccine*, published in 2020:

The trial enrolled 30,420 volunteers who were randomly assigned in a 1:1 ratio to receive either vaccine or placebo (15,210 participants in each group). More than 96% of participants received both injections, and 2.2% had evidence (serologic, virologic, or both) of SARS-CoV-2 infection at baseline. Symptomatic COVID-19 illness was confirmed in 185 participants in the placebo group (56.5 per 1,000 person-years; 95% confidence interval (CI), 48.7 to 65.3) and in 11 participants in the mRNA-1273 group (3.3 per 1,000 person-years; 95% CI, 1.7 to 6.0); vaccine efficacy was 94.1% (95% CI, 89.3 to 96.8%; $P < 0.001$).

Here is my take on how to interpret these results in the light of the concepts you've been learning. Keep in mind that I'm no health expert and my commentary is purely about the statistical and causal inference concepts.

First, they defined the treatment and control (placebo) groups, saying that the treatment was randomly assigned, which ensures independence of the treatment from the potential outcomes. This would allow them to identify the causal effect of the vaccine from statistical the quantities $E[Y|T = 0]$ and $E[Y|T = 1]$. Next, they define the outcome as being the presence of symptomatic COVID-19 per 1,000 person-years. Finally, they report the 95% CI for the estimate

of $E[Y|T = 0]$ and $E[Y|T = 1]$ as being 48.7 to 65.3 and 1.7 to 6.0, respectively. This tells you that symptomatic COVID-19 was detected far less in those with the vaccine, compared to those that got the placebo. They report the efficacy of the vaccine, $E[Y|T = 0]/E[Y|T = 1]$, as well as the 95% confidence interval around it, 89.3 to 96.8%.

One final word of caution here. Confidence intervals are trickier to interpret than at first glance. For instance, I *shouldn't* say that a particular 95% confidence interval contains the true mean with 95% chance. In frequentist statistics, the population mean is regarded as a true population constant. This constant is either inside or outside a particular confidence interval. In other words, a specific confidence interval either contains or doesn't contain the true mean. If it does, the chance of containing it would be 100%, not 95%. If it doesn't, the chance would be 0%. Instead, in confidence intervals, the 95% refers to the frequency that such confidence intervals, computed in many studies, contains the true mean. The 95% is our confidence in the algorithm used to calculate the 95% CI, not on the particular interval itself.

Now, having said that, as an economist (statisticians, please look away now), I think this purism is not very useful. In practice, you will see people saying that the particular confidence interval contains the true mean 95% of the time. Although wrong, this is not very harmful, as it still places a visual degree of uncertainty in your estimates. What I mean by this is that I would rather you have a confidence interval around your estimate and interpret it wrong than avoid the confidence interval in fear of misinterpretation. I don't care if you say

they contain the true mean 95% of the time. Just, please, never forget to place them around your estimates; otherwise, you will look silly.

CREDIBLE INTERVALS

If you really want to attach a probability statement to a parameter estimate being inside an interval, you should check Bayesian credible intervals. However, from my experience, in most situations (especially when the sample size is relatively large) they tend to yield something similar to the frequentists confidence interval. This is also why I tend to be more forgiving of misinterpretation of the confidence interval.

Hypothesis Testing

Another way to incorporate uncertainty is to state a hypothesis test: is the difference in means between two groups statistically different from zero (or any other value)? To answer these types of questions, you need to recall that the sum or difference of two independent normal distributions is also a normal distribution. The resulting mean will be the sum or difference between the two distributions, while the variance will always be the sum of the variances:

$$N(\mu_1, \sigma_1^2) - N(\mu_2, \sigma_2^2) = N(\mu_1 - \mu_2, \sigma_1^2 + \sigma_2^2)$$

$$N(\mu_1, \sigma_1^2) + N(\mu_2, \sigma_2^2) = N(\mu_1 + \mu_2, \sigma_1^2 + \sigma_2^2)$$

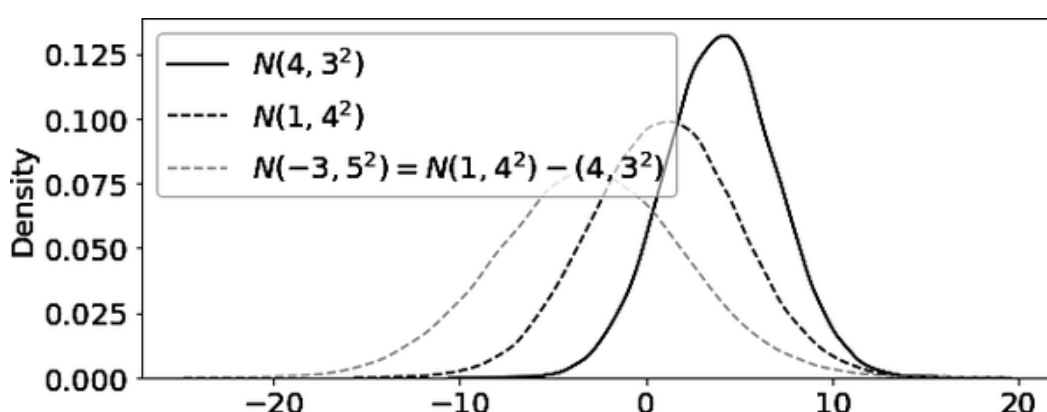
If you don't remember, it's OK. You can always use code and simulated data to check it for yourself:

```
In [13]: import seaborn as sns
         from matplotlib import pyplot as plt

         np.random.seed(123)

         n1 = np.random.normal(4, 3, 30000)
         n2 = np.random.normal(1, 4, 30000)
         n_diff = n2 - n1
```

```
plt.figure(figsize=(10,4))
sns.distplot(n1, hist=False, label="$N(4, 3^2)$")
sns.distplot(n2, hist=False, label="$N(1, 4^2)$")
sns.distplot(n_diff, hist=False,
             label=f"$N(-3, 5^2) = N(1, 4^2) - N(4, 3^2)$")
plt.legend();
```



If you take two groups, each with a distribution attached to it, and subtract one from the other, you'll end up with a third distribution. The mean of this final distribution will be the difference in the means, and the standard deviation will be the square root of the sum of the variances. Since you are talking about the distributions of experiment averages, you can think about the standard deviation of these as the standard error of the mean:

$$\mu_{diff} = \mu_1 - \mu_2$$

$$SE_{diff} = \sqrt{SE_1^2 + SE_2^2}$$

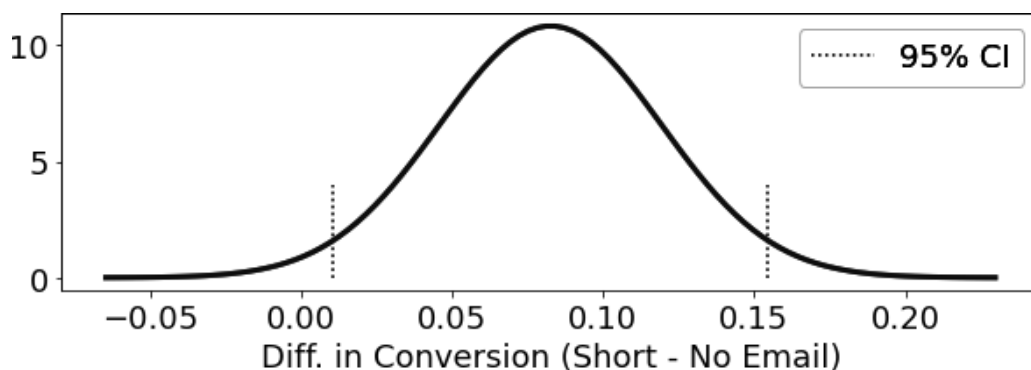
You can use this idea in the problem of comparing the conversion from your cross-sell email experiment. If you take the estimated distribution of two groups—let's say, the short email and the no email group—and subtract one from the other, you get the distribution of

the difference. With this distribution, you can easily construct a 95% confidence interval for the difference in means:

```
In [14]: diff_mu = short_email.mean() - no_email.mean()
diff_se = np.sqrt(no_email.sem()**2 + short_email.sem()**2)

ci = (diff_mu - 1.96*diff_se, diff_mu + 1.96*diff_se)
print(f"95% CI for the difference (short email - no email) is {ci}")
```

```
Out[14]: 95% CI for the difference (short email - no email) is
(0.01023980847439844, 0.15465380854687816)
```



Null Hypothesis

With this interval, you can answer questions about what is called a *null hypothesis*. For example, you can state the hypothesis that there is no difference in conversion rate between a short email and no email at all. You'll usually use H_0 to talk about the null hypothesis:

$$H_0 : Conversion_{no_email} = Conversion_{short_email}$$

Once you have this hypothesis, it's time to ask yourself, "Is it likely that I would observe such a difference if the null hypothesis were true?" You'll look at the data and see if it conforms to your null hypothesis. If it doesn't, you'll say that seeing such data would be too weird, if the null hypothesis were true, and hence you should reject it. One way to do this is with the confidence intervals you have just constructed.

Notice how the preceding 95% confidence interval does not contain zero. Also, recall that this is the CI of the difference between conversion rates. Since the null hypothesis states that this difference is zero, but you can see that the confidence interval is entirely outside zero, you can say that the probability of seeing such a result would be too low, if the null hypothesis were true. Hence, you can reject the null hypothesis with 95% confidence.

SIGNIFICANCE LEVEL

The significance level, α , is the chance of rejecting the null when it is true—committing a Type I error. Significance is set prior to gathering or analyzing the data. To achieve a certain significance level, say, 5%, you would construct a $1 - \alpha$ confidence interval, say 95%, around your estimate during the analysis.

Of course you can also formulate other null hypotheses, besides the one that states no difference at all. For example, let's say there is some cost to sending emails, which is very realistic. Even if there is no significant monetary cost, if you send too many emails to customers, eventually they will flag you as spammers, which will shut down this communication channel with them, leading to lower sales in the future. Under this situation, perhaps the marketing team is only willing to roll out the cross-sell email if the lift in conversion rate is higher than 1%. Then, you can state the null hypothesis as follows: "the difference in conversion rate is 1%." To test this hypothesis, all you need to do is shift the confidence interval by subtracting 1% from the difference in means:


```
In [15]: # shifting the CI
diff_mu_shifted = short_email.mean() - no_email.mean()
diff_se = np.sqrt(no_email.sem()**2 + short_email.sem()**2)

ci = (diff_mu_shifted - 1.96*diff_se, diff_mu_shifted + 1.96*diff_se)
print(f"95% CI 1% difference between (short email - no email) is {ci}")
```

```
Out[15]: 95% CI 1% difference between (short email - no email) is
(0.00023980847439844521, 0.14465380854687815)
```

Since this 95% CI is also above zero, you can also reject this other null hypothesis. However, now the 95% CI is very close to zero, meaning you would not be able to reject the null hypothesis of the effect being equal to something like 2%, at least not with a 95% confidence.

NONINFERIORITY TESTING

In this book, most null hypotheses will be stated as an equality (usually to zero). This type of null is motivated by the desire to treat only if an effect is found to be significantly different from zero.

However, in some situations, you want to act only if a treatment effect is equal to zero. Consider, for instance, the case where you want to shut down a marketing campaign. You only want to do that if its effect is negligible (or not high enough to compensate its costs). In these situations, *you want to state the null in terms of a parameter being different from some value.*

That's because not being able to reject a null like $H_0 = 0$ is not the same thing as accepting it as true. This is known as the famous adage that “absence of evidence is not evidence of absence.” The $H_0 = 0$ could be rejected simply because the sample size is too small, yielding a large confidence interval. This, however, doesn't point in the direction of it being true.

To work around this issue, statisticians created noninferiority testing, which is a way to test for a treatment being equal to another (or having a zero treatment effect). The basic idea is to see if the confidence interval contains zero, while also making sure it is small enough.

Test Statistic

Besides confidence intervals, sometimes it is useful to think about rejecting the null hypothesis in terms of a *test statistic*. These statistics are often constructed such that higher values point toward rejection of the null. One of the most commonly used test statistics is the t-

statistic. It can be defined by normalizing the distribution that gives rise to the confidence interval:

$$t_{\Delta} = \frac{\mu_{\Delta} - H_0}{SE_{\Delta}} = \frac{(\mu_1 - \mu_2) - H_0}{\sqrt{\sigma_1^2/n_1 + \sigma_2^2/n_2}},$$

where H_0 is the value defined by your null hypothesis.

Notice how the numerator is simply the difference between the observed average difference and the null hypothesis. If the null were true, the expected value of this numerator would be zero:

$E[\mu_{\Delta} - H_0] = 0$. The denominator is simply the standard error, which normalizes the statistic to have unit variance. It ensures that t_{Δ} follows a standard normal distribution— $N(0, 1)$ —if the null is true. Since t_{Δ} is centered around zero under the null, values above or below the 1.96 would be extremely unlikely (appear less than 95% of the time). This means you can also reject the null hypothesis if you see such an extreme t-statistics. In our running example, the statistics associated with H_0 of no effect is greater than 2, meaning you can reject it at a 95% confidence level:

```
In [16]: t_stat = (diff_mu - 0) / diff_se
          t_stat
```

```
Out[16]: 2.2379512318715364
```

Additionally, since t-statistic is normally distributed under the null, you can use it to easily compute p-values.

T VERSUS NORMAL

Technically speaking using the normal distribution here is not accurate. Instead, you should use the T distribution with degrees of freedom equal to the sample size minus the number of parameters you've estimated (2, since you are comparing two means). However, with samples above 100, the distinction between the two is of little practical importance.

p-values

Previously, I've said that there is less than a 5% chance you would observe such an extreme difference if the conversion of customers that received no email and short email were the same. But can you precisely estimate what that chance is? How likely are you to observe such an extreme value? Enter p-values!

Like with confidence intervals (and most frequentist statistics, as a matter of fact), the true definition of p-values can be very confusing. So, to not take any risks, I'll copy the definition from Wikipedia: "the p-value is the probability of obtaining test results at least as extreme as the results actually observed during the test, assuming that the null hypothesis is correct."

To put it more succinctly, the p-value is the probability of seeing such data, if the null hypothesis were true (see [Figure 2-4](#)). It measures how unlikely that measurement you are seeing is, considering that the null hypothesis is true. Naturally, this often gets confused with the probability of the null hypothesis being true. Note the difference here. The p-value is *not* $P(H_0|data)$, but rather $P(data|H_0)$.

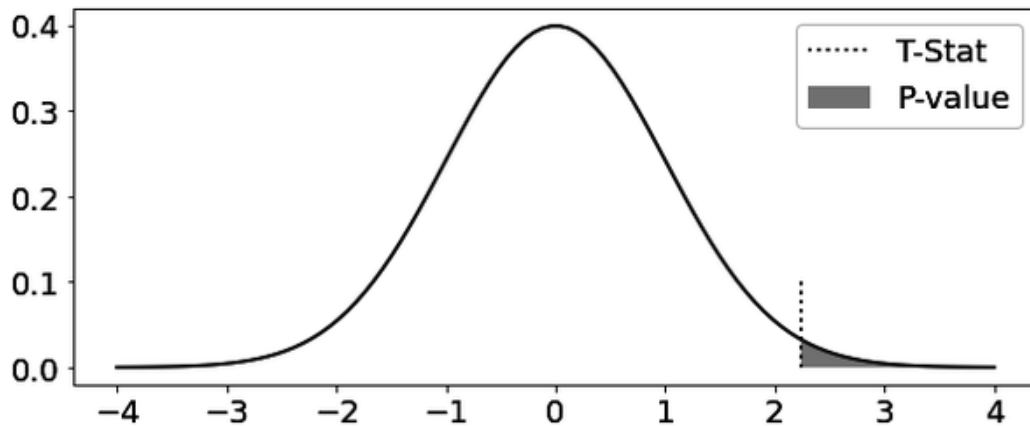


Figure 2-4. p-value is the probability of seeing a extreme statistic, given that the null hypothesis is true

To get the p-value, all you need to do is compute the area under the standard normal distribution before the test-statistic for a one-sided null hypothesis (“the difference is greater than x” or “the difference is smaller than x”) and multiply the result by 2 for a two-sided null hypothesis (“the difference is x”):

```
In [17]: print("p-value:", (1 - stats.norm.cdf(t_stat))
```

```
Out[17]: P-value: 0.025224235562152142
```

The p-value is interesting because it frees you from having to specify a confidence level, like 95% or 99%. But, if you wish to report one, from the p-value, you know precisely at which confidence your test will pass or fail. For instance, with a p-value of 0.025, you’ll have significance up to the 2.5% level. So, while the 95% CI for the difference will not contain zero, the 99% CI will. This p-value also means that there is only a 2.5% chance of observing this extreme test statistic, if the difference was truly zero.

FACE-TO-FACE VERSUS ONLINE LEARNING

Besides the direct impact of the virus, the 2020 pandemic brought other important issues with it. Chief among those was the fact that kids could not go to school, so learning was taken to the online environment for as much as two years. It's hard to estimate the generational impact this will have, since the decision to come back from the online environment to a face-to-face one was not randomized. In Brazil, for example, public schools took longer to open up, compared to private schools.

However, although certainly not trivial, one can design an experiment to test the impact of online versus face-to-face learning, as Figlio, Rush, and Yin did in "Is It Live or Is It Internet? Experimental Estimates of the Effects of Online Instruction on Student Learning" (2013). Here is the abstract:

Students in a large introductory microeconomics course at a major research university were randomly assigned to live lectures versus watching these same lectures in an internet setting, where all other factors (e.g., instruction, supplemental materials) were the same. Counter to the conclusions drawn by a recent U.S. Department of Education meta-analysis of nonexperimental analyses of internet instruction in higher education, we find modest evidence that live-only instruction dominates internet instruction. These results are particularly strong for Hispanic students, male students, and lower-achieving students. We also provide suggestions for future experimentation in other settings.

Notice that this study was conducted at a university in the US; it's hard to say those results will generalize to basic education and to other countries. In technical terms, we say that the study has internal

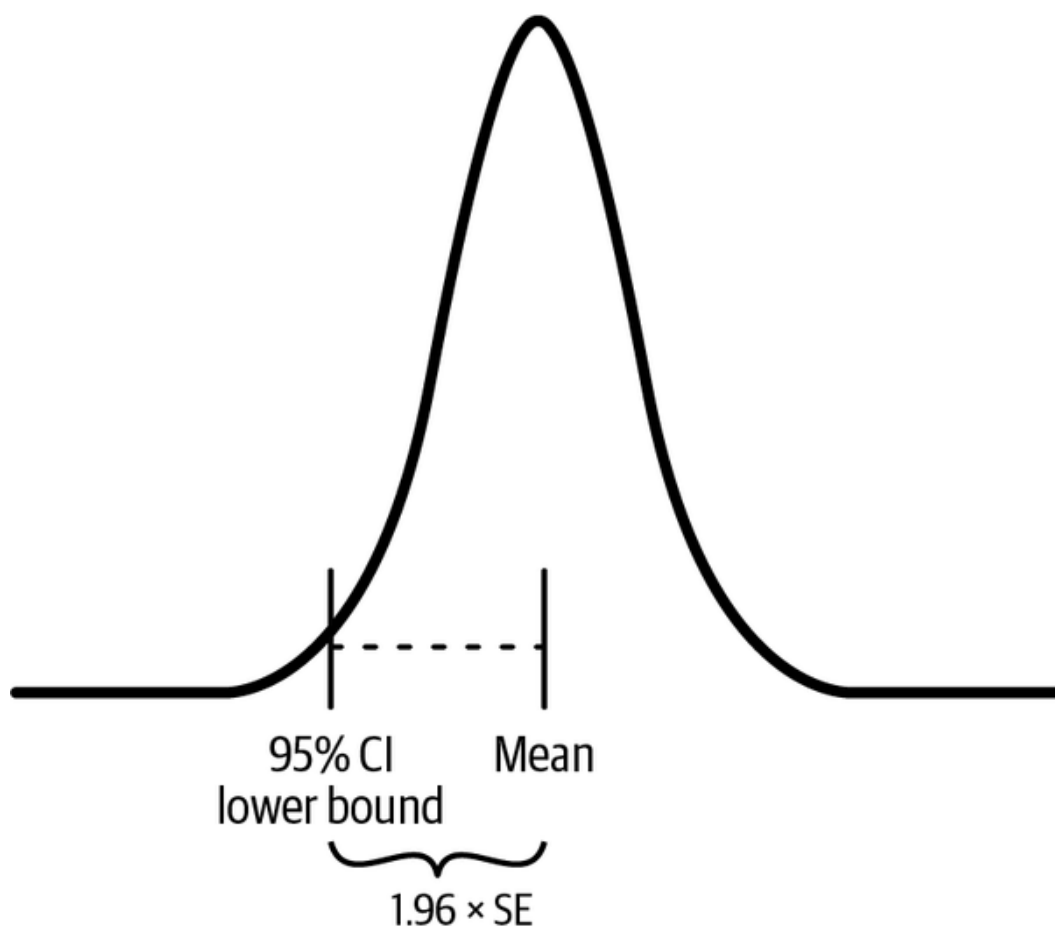
validity, as treatment and control groups are comparable due to randomization. But this study might not have external validity in terms of generalizing its results to other settings, since the people in it were not a random sample of the population, but rather economics students from a US university.

Power

So far, you've been looking into these statistical concepts from the perspective of a data analyst who's been presented with the data from an existing test. You are treating the data as given. But what if you are asked to design an experiment, instead of just reading one that was already designed? In this case, you need to decide the sample you would like to have for each variant. For example, what if you haven't yet run the cross-sell email experiment, but, instead, need to decide how many customers you should send the long email and how many, the short email and no email at all? From this perspective, the goal is to have a big enough sample so that you can correctly reject the null hypothesis of no effect, if it is indeed false. *The probability that a test correctly rejects the null hypothesis is called the power of the test.* It's not only a useful concept if you want to figure out the sample size you need for an experiment, but also for detecting issues in poorly run experiments.

Power is closely related to statistical significance. While α is the chance of rejecting the null hypothesis when it is actually true, power ($1 - \beta$) is the chance of rejecting the null when it is false. In some sense, power is also defined in terms of α as in order to correctly reject the null, you need to specify how much evidence you need for rejection.

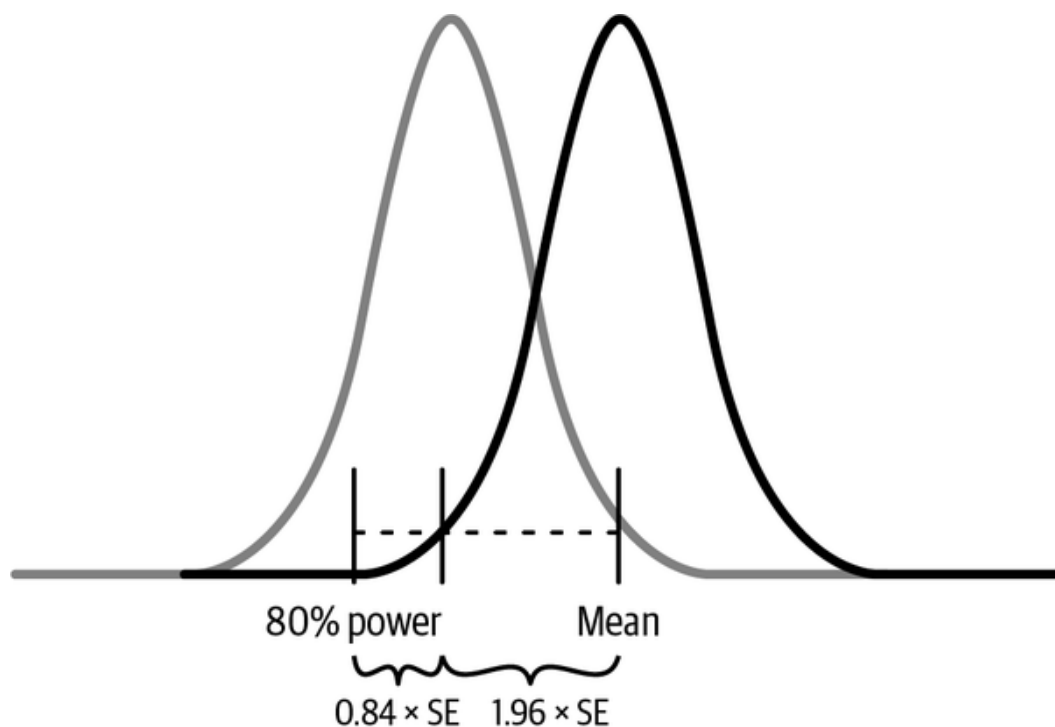
Recall how the 95% confidence interval means that 95% of the experiments will contain the true parameter you are trying to estimate. This also means that 5% of them won't, which will cause you to falsely reject the null hypothesis 5% of the time. With $\alpha = 0.05$, you need δ , the difference between the parameter estimate and the null hypothesis, to be at least $1.96 SE$ away from zero in order to conclude that it is statistically significant. That's because $\delta - 1.96 SE$ is the lower end of the 95% confidence interval.



OK, so you need $\delta - 1.96 SE > 0$ to claim the result as significant. But how likely are you to see this significant difference? This is where you need to think about power. Power is the chance of correctly rejecting the null, or $1 - \beta$, with β being the probability of not rejecting the null when it is false (probability of false negative). The

industry standard for power is 80%, meaning that you'll only have a 20% ($\beta = 0.2$) chance of not rejecting the null, when it is indeed false. To achieve 80% power, you need to reject the null hypothesis 80% of the time when it is false. Since rejecting the null means that $\delta - 1.96 SE > 0$, you need to get this big difference 80% of the time. In other words, you need to get the lower end of the 95% CI above zero, 80% of the time.

What is striking (or not) is that the lower end of the 95% confidence interval also follows a normal distribution. Just like the distribution of the sample average, the distribution of the lower end of the 95% CI has variance equal to the SE , but now the mean is $\delta - 1.96 SE$. It is just the distribution of the sample average, shifted by $1.96 SE$. Therefore, in order to have $\delta - 1.96 SE > 0$ 80% of the time (80% power), you need the difference to be $1.96 + 0.84 SE$ away from zero: 1.96 to give you the 95% CI and 0.84 so that the lower end of that interval falls above zero 80% of the time.



```
In [18]: stats.norm.cdf(0.84)
```

```
Out[18]: 0.7995458067395503
```

Sample Size Calculation

Another way to look at this is to realize that δ —the difference between the null hypothesis and the observed estimate—must be detectable if the null is false. With $\alpha = 5\%$ and $1 - \beta = 80\%$, the detectable effect is given by $2.8 SE = (1.96 SE + 0.84 SE)$. So, if you want to craft a cross-sell email experiment where you want to detect a 1% difference, you must have a sample size that gives you at least $1\% = 2.8 SE$. If you open up the SE formula for the difference, you have

$SE_{\Delta} = \sqrt{SE_1^2 + SE_2^2}$. But recall that you are now speaking from the perspective of an analyst who has not seen the experiment, but is actually trying to design it. In this case, you don't have the SE of the treated group, but you can assume that the variance in both treated and control will be the same and hence $SE_{\Delta} = \sqrt{2SE^2} = \sqrt{2\sigma^2/n} = \sigma\sqrt{2/n}$. Plugging this in the detectable difference, you end up with a pretty simple formula for determining the sample size of each variant in a test if you want 80% power and 95% significance:

$$\delta = 2.8\sigma\sqrt{2/n}$$

$$n = 2 * 2.8^2 \sigma^2 / \delta^2 \approx 16\sigma^2 / \delta^2$$

where δ is the detectable difference and I've rounded $2 * 2.8^2$ to be conservative. Applying this formula to your data, using the variance of the control group as our best guess for σ^2 , you end up with the following required sample size:

```
In [19]: np.ceil(16 * no_email.std() ** 2 / 0.01)
```

```
Out[19]: 66.0
```

```
In [20]: data.groupby("cross_sell_email").size()
```

```
Out[20]: cross_sell_email
         long          109
         no_email       94
         short          120
         dtype: int64
```

This is of course invaluable in terms of experiment design, but is also good news for the cross-sell experiment we currently have. In it, we have more than 100 samples for both treatment groups and 94 samples for the control, which indicates a properly powered test.

SEE ALSO

This very simple way of calculating sample size was taken from “A/B Testing Intuition Busters: Common Misunderstandings in Online Controlled Experiments” (2022), by Ron Kohavi et al. This sample size formula is only one of the many very interesting and useful things presented in the article, so I definitely recommend you check it out.

Key Ideas

The idea of this chapter was to link causal identification with estimation (and also review some important statistical concepts). Recall that the goal of causal inference is to learn about causal quantities from data. The first step in the process is identification, where you use key assumptions to go from unobservable causal quantities to observable statistical quantities you can estimate from data.

For example, the ATE is a causal quantity; it is defined by the unobservable potential outcomes $ATE = E[Y_1 - Y_0]$. To identify the ATE, you use the independence assumption, $T \perp (Y_0, Y_1)$, which allows you to write it in terms of observable quantities, $E[Y|T = 1]$, and $E[Y|T = 0]$. That is, under the independence assumption:

$$E[Y_1 - Y_0] = E[Y|T = 1] - E[Y|T = 0]$$

You also saw how you could use randomized control trials (RCTs) to make this assumption more plausible. If you randomize the treatment, you are brute-forcing it to be independent from the potential outcomes Y_t .

But identification is just the first step in causal inference. Once you are able to write the causal quantities in terms of statistical quantities, you still need to estimate those statistical quantities. For instance, even though you can write the ATE in terms of $E[Y|T = 1]$ and $E[Y|T = 0]$, you still need to estimate them.

The second part of this chapter covered statistical concepts used in that estimation process. Specifically, you learned about the standard error:

$$SE = \sigma / \sqrt{n},$$

and how to use it to place confidence intervals around an estimate μ :

$$\hat{\mu} \pm z * SE,$$

where z is the value between which lies $\alpha\%$ of the mass of a normal distribution.

You also learned how to construct a confidence interval for a difference in averages between two groups, which boiled down to summing the variances for those groups and finding a standard error for the difference:

$$SE_{diff} = \sqrt{SE_1^2 + SE_2^2}$$

Finally, you learned about power and how it can be used to calculate the sample size for an experiment you wish to run. Specifically, for 95% confidence and 80% power, you could simplify the sample size formula to:

$$N = 16 * \sigma^2 / \delta$$

where σ^2 is the variance of the outcome and δ is the detectable difference.